No Cache vs Cache Table

Program	No Cache - cycles	Cache - cycles
A	240096	90044
B	480096	60052
C	3680280	460203
D	3680256	460192
E	800096	310027

Why?

Program A is just a sub r9, r8 10,000 times which is the same instruction over and over again so it makes sense that the cache would be able to lower the cycle counter much lower. But it still hits the cache a lot.

Program B uses a loop to accomplish the same thing as Program A and makes use of the same instructions over and over again.

Program C does a bunch of math on some large integers. It uses different operations and so the cache would not be as effective, but since we have so many cache lines it is able to make use of all the instructions.

Program D This is the same thing as Program C

Program E is almost the same as A and B but makes use of the ildb and so it has more accesses to memory than the other 2.